

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB

**WEEK 06 TASK: BACKEND DEVELOPMENT & DATABASE
IMPLEMENTATION PROGRESS REPORT**

TEAM NAME: CSE4204-8D-T01 || SECTION: 8D

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Section: 8D || **Team ID:** CSE4204-8D-T01

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

1. Team Information & Core Engineering Roles

Full Member Name	Student ID	System Design & Engineering Responsibilities
Bashir Alam	11220320963	Team Leader. Core backend system architecture, API design, Celery worker workflow orchestration, Open Router API & ML models integration.
Tisha Rani Chakroborty	11220320954	Frontend Developer. UI/UX layout design systems, interactive dashboard wireframes, and responsive views matching via Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL schema structure planning, entity relationship mapping, and relational table indexing within Supabase.
Ratul Raha Shuvo	11220320949	Backend Support. Payment gateway integrations sub-system design, transactional report templates, and REST API test suite architecture.

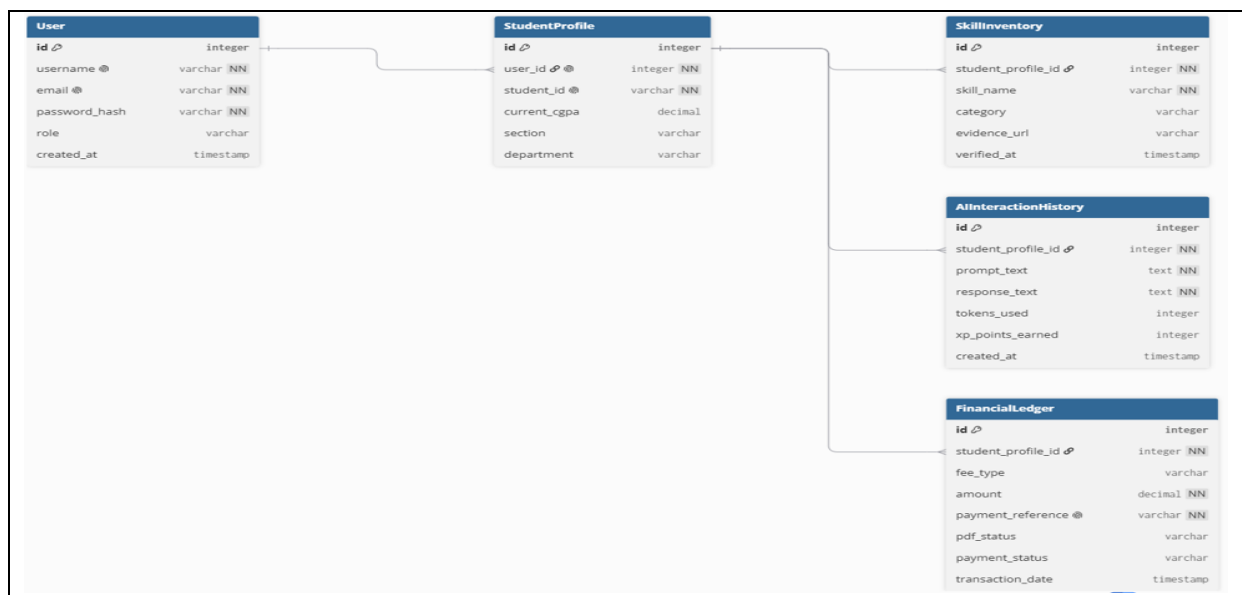
2. Database Architecture & Schema Implementation

The cloud database instance has been configured and tables have been successfully materialized on the Supabase PostgreSQL platform based on the approved ER Diagram structure. The structural layout includes explicit foreign keys, primary keys, and constraint validations:

2.1 Implemented Relational Tables

1. **User Table (Authentication Core):** Manages essential access credentials, roles (Student, Faculty, Admin, Super Admin), and encrypted password blocks.
2. **Student Profile Table:** Keeps track of vital academic data including institutional Student ID registries, GPA scores, and links to user credentials.
3. **Skill Inventory Table:** Acts as an immutable ledger storing practical engineering skills tagged against individual student records.
4. **AI Interaction History Table:** Archives detailed token usage, system chat logs, and contextual context strings from the AI Mentor Hub.
5. **Financial Ledger / Transaction Table:** Logs digital payment gateway reference numbers, enrollment validation tags, and automated accounting workflows.

Database Architecture



User		StudentProfile	
id 🔗	integer	id 🔗	integer
username 🔒	varchar NN	user_id 🔗 🔒	integer NN
email 🔒	varchar NN	student_id 🔒	varchar NN
password_hash	varchar NN	current_cgpa	decimal
role	varchar	section	varchar
created_at	timestamp	department	varchar

3. Authentication & Authorization Workflow

To safeguard corporate academic identities, a highly secure JWT-based stateless framework was implemented:

1. **Registration:** The client applications send payloads containing the user credentials to the backend. The application layer interceptor hashes the cleartext passwords using secure PBKDF2 blocks before saving them to the user schema.
2. **Login Verification:** Upon sending active credentials to the server, the backend system evaluates the cryptographic hash values. If authenticated, the service issues an isolated, time-bound **Access Token** and a long-lived **Refresh Token**.
3. **Route Protection:** Protected API endpoints (e.g., logging skills or launching mentor chats) require a valid Bearer <Token> string embedded directly into the standard incoming HTTP Authorization requests Header.

Backend Folder Structure:

```

backend_root/
|
|— .env                # System Environment Variables (API Keys, DB Credentials)
|— db.sqlite3          # Local Development Database Instance
|— manage.py           # Django Management CLI Utility
|— requirements.txt     # Project Python Dependencies & Packages
|
|— backend/            # Core Project Configuration Directory
|   |— __init__.py
|   |— asgi.py          # Asynchronous Server Gateway Interface

```

```

| | — settings.py      # Main Project Configurations & Installed Apps
| | — urls.py         # Global URL Routing & Endpoint Mapping
| | — wsgi.py         # Web Server Gateway Interface Configuration
|
| — portal/          # Main Architecture Application Module
|   | — __init__.py
|   | — admin.py      # Django Admin Panel Models Registration
|   | — apps.py       # Application Context Configuration
|   | — context_processors.py # Custom Global Template Context Injectors
|   | — models.py     # Database Schema Blueprint Definitions
|   | — tests.py      # Backend Automation Test Scripts
|   | — urls.py       # Portal-Specific Endpoint Routing Matrices
|   | — utils.py      # AI Inference Engine & Helper Utilities
|   | — views.py      # Core Controller Logic & Request Handlers
|   |
|   | — migrations/   # Database Schema Version Control Logs
|   |   | — 0001_initial.py # Automated Supabase/SQL Database Migrations
|   |   | — __init__.py
|   |
|   | — static/       # Static Asset Repositories
|   |   | — css/
|   |   |   | — output.css # Compiled Tailwind CSS Utility Layer
|   |   |   | — js/
|   |   |     | — main.js  # Client-Side Interactive Scripts
|   |
|   | — templates/    # Server-Side Rendered HTML Layer
|   |   | — academic_matrix.html # Semester Course Mapping Interface
|   |   | — admin_dashboard.html # Superuser Management Control Center
|   |   | — base.html      # Main Global Blueprint Layout Shell
|   |   | — chat.html     # Real-Time AI Copilot Interaction Console
|   |   | — home.html     # Institutional ERP Entry Landing Page
|   |   | — login.html    # Secure User Authentication Gateway
|   |   | — student_profile.html # Dynamic Skill Ledger & Portfolio View

```

